

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

[Introduction to computer Graphics]

create, store, manipulate
Image } computer Graphics

Data extraction → computer vision

Img. processing → making changes to img.

create, store, manipulation → computer graphics

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

[Graphics Computation]

$$\text{Total pixels} = \text{width} \times \text{Height}$$

$$= 1920 \times 1080$$

$$= 2073600 \text{ pixels per frame}$$

$$\text{Total pixels per second} = \text{pixels per frame}$$

$$\times 60 \text{ FPS pixels per second}$$

$$= 2073600 \times 60$$

$$\text{Aspect Ratio} = \frac{\text{width}}{\text{Height}} = \frac{1920}{1080} = 16:9$$

$$\text{color depth} = \text{bits per pixel}$$

$$\text{let, bit} = 24 \Rightarrow 2^n, 2^{24} \text{ of combinations}$$

$$\text{Image size} = \text{width} \times \text{Height} \times \text{Depth (bits per pixel)}$$

$$= 1920 \times 1080 \times 24 \text{ bits}$$

$$\text{In bytes} = \frac{\text{Image size}}{8} = \frac{1920 \times 1080 \times 24}{8}$$

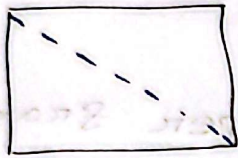
TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

$$\text{Pixel per Inch (PPI)} = \frac{\sqrt{\text{width}^2 + \text{Height}^2}}{\text{Diagonal size (in Inches)}}$$



$$\text{Time per frame} = \frac{1}{\text{FPS}}$$

$$\text{pixels "x"x"y" = } N \times 16.67$$

#

$$\frac{1920 \times 1080 \times 4}{8} = \frac{\text{width} \times \text{height} \times \text{bits per pixel}}{\text{bytes}} \quad N \times \frac{1}{120}$$

$$\text{Total pixels per second} = \frac{1920 \times 1080 \times 4 \times 120}{8}$$

$$\text{Img size} = \frac{1920 \times 1080 \times 4 \times 24}{8}$$

GOOD LUCK

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

(Anti aliasing)

Pixel

Either colored / uncolored } Aliasing

- 1) Unweighted Area Sampling
- 2) weighted Area Sampling

$$I_{x,y} = I_{max} \cdot \Delta A \cdot \text{weight}$$

↳ Area overlap for pixels

TOPIC NAME : _____

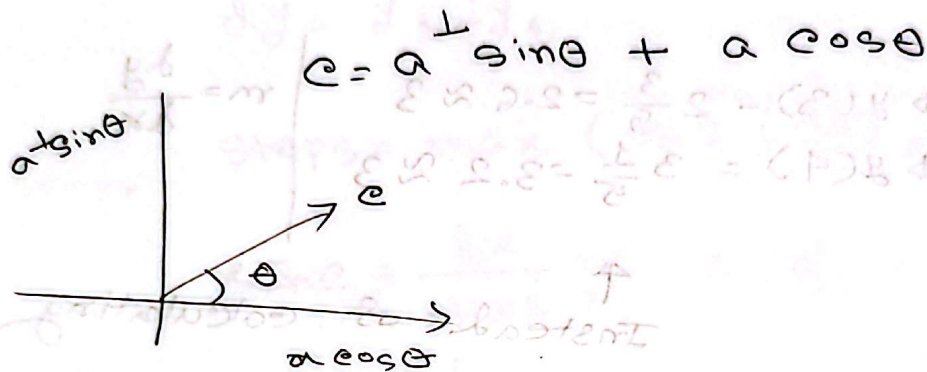
DAY : _____

TIME : _____

DATE : / /

[Vectors and Line Drawing Introduction]

Rotation in 2D:



$$a = (a_x, a_y)$$

$$a^{\perp} = (-a_y, a_x)$$

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: 30/06/2025

[Scan Conversion]

$y = mx + b \rightarrow$ takes too much time

Pixel

$$(3,3) \rightarrow y(3) = 2 \frac{3}{5} = 2.6 \approx 3$$

$$(4,3) \rightarrow y(4) = 3 \frac{1}{5} = 3.2 \approx 3$$

$$m = \frac{dy}{dx}$$

↑
Instead of calculating multiple times we can calculate the difference initially.

{ Digital
Differential Analyzer }

DDA:

$$y_{k+1} = y_k + m$$

$$x_{k+1} = x_k + 1$$

} slope: between
-1 and 1

$$y_{k+1} = y_k + 1$$

$$x_{k+1} = x_k + \frac{1}{m}$$

} else

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

DDA (x_0, y_0, x_1, y_1) {

$$\Delta x = x_1 - x_0$$

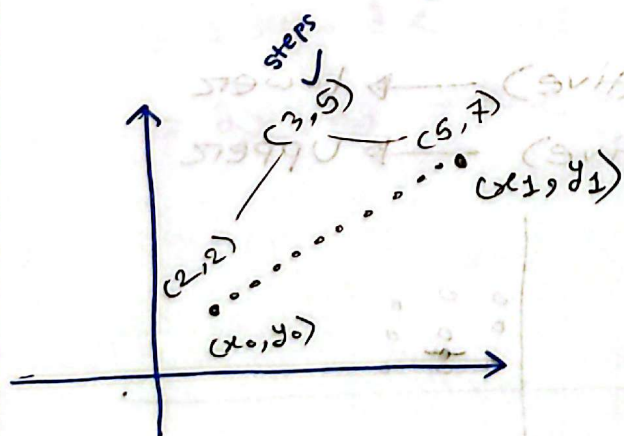
$$\Delta y = y_1 - y_0$$

$$\text{steps} = \max(\text{abs}(\Delta x), \text{abs}(\Delta y));$$

$$x_{\text{Inc}} = \frac{\Delta x}{\text{steps}};$$

$$y_{\text{Inc}} = \frac{\Delta y}{\text{steps}};$$

for ($i = 1, i \leq \text{steps}; i++$) {
 putpixels ($\text{round}(x_0), \text{round}(y_0)$);
 $x_0 = x_0 + x_{\text{Inc}};$
 $y_0 = y_0 + y_{\text{Inc}};$



∴ no multiplication

but, still round off

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

midpoint Line :

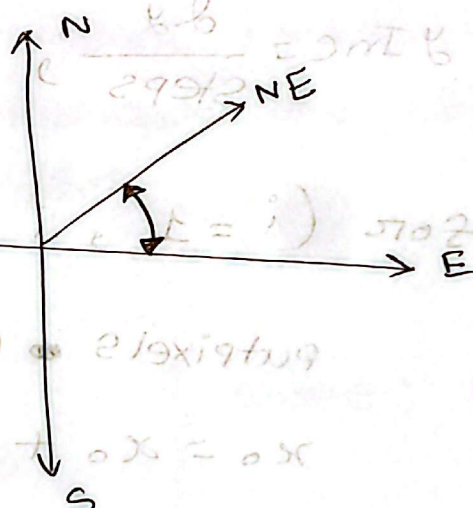
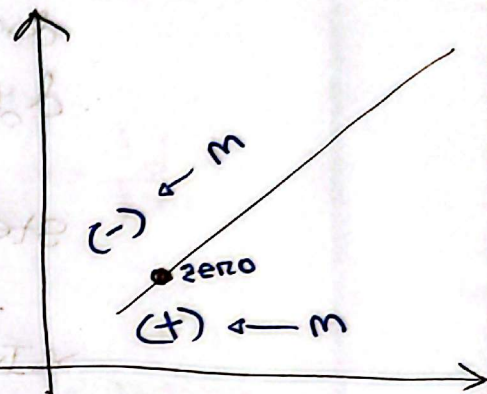
$$f(x, y) = x + y + 1 = 0$$

$$f(2, 3) = 2 + 3 + 1 = 6$$

$$f(-2, 3) = -2 + 3 + 1 = 2$$

$$f(-7, 3) = -7 + 3 + 1 = -3$$

$$f(-1, 1) = -1 + 1 + 1 = 1$$



(0, negative) → lower
(0, positive) → upper

$0 < \Delta E \rightarrow \Delta + \Delta y - \Delta x$
 $0 > \Delta E \rightarrow \Delta + \Delta y$



TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Decision parameter = $d = ax_p + by_p + c + a + 0.5b$

↓

$d_{new} = ax_p + by_p + c + 2a + 1.5b$

which pixel is to choose next

$= ax_p + by_p + c + a + 0.5b + a + b$

$= d + a + b$

NE Inc.

$d_{new} = d + a$
E Inc.

$d = d_{init}$
 $= dy - 0.5dx$
Still, round-off

we can do this,

$d = d_{init}$

$= dy - dx$

$= dy + dx - dx$

$a = dy$

$b = -dx$

$c =$

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

Find out the first 7 pixels of the line segment starting from (0,2) to (70,52) using midpoint line algorithm.

$\Rightarrow dx = 70 - 0 = 70$
 $dy = 52 - 2 = 50$

$NE Inc = 2dy - 2dx = -40$
 $E Inc = 2dy = 100$

$\nearrow d - 40$
 $\searrow d + 100$

x	y	d	NE (+1,+1) / E (+1,0)	d updating	Pixel
0	2	30 > 0	NE	$30 * 2 * 50 - 2 * 70 = -10$	(0,2)
1	3	-10 < 0	E	90	(1,3)
2	3	90 > 0	NE	$90 - 40 = 50$	(2,3)
3	4	50 > 0	NE	$50 - 40 = 10$	(3,4)
4	5	10 > 0	NE	$10 - 40 = -30$	(4,5)
5	6	-30 < 0	E	$-30 + 100 = 70$	(5,6)
6	6	70 > 0	NE	$70 - 40 = 30$	(6,6)

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

$$\Delta = \Delta_{init}$$

$$= 2\Delta y - \Delta x$$

$$= 2.50 - 70$$

$$= 30$$

$$\Delta > 0$$

yes

NE

no

$$\Delta \leftarrow \Delta + 2\Delta y - 2\Delta x$$

$$= 30 + 2.50 - 2.70$$

$$= -10$$

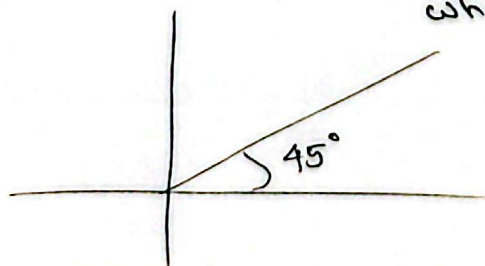
$$\Delta \leftarrow \Delta + 2\Delta y$$

$$= 30 + 2.50$$

$$= -10 + 100$$

$$= 90$$

we can only draw in 1st quadrant
where, $\theta = 45^\circ$



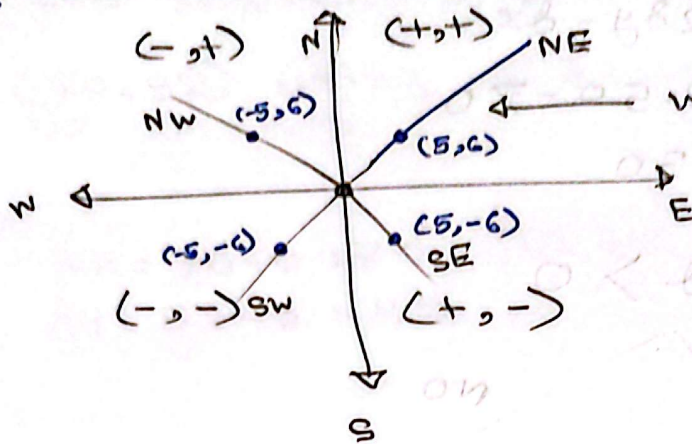
TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

8 way symmetry:



we have already done this

but what happens for other angles?

Since, ^{for} multiple angles we just can't introduce new formulas.

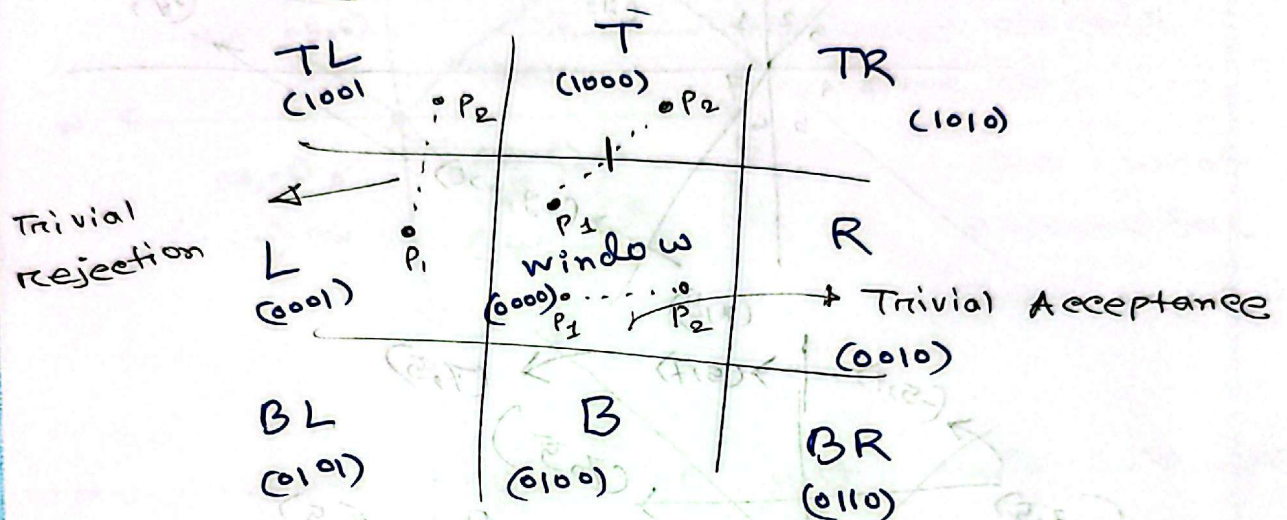
TOPIC NAME : _____

DAY : _____

TIME : _____ DATE: 07/06/2025

[Clipping]

(Cohen - Sutherland Algo)



Outcode	→	T	B	R	L
00		1	0	1	0
		1	0	0	0
		1	0	0	1
		0	0	0	1
		0	1	0	1
		0	1	0	0
		0	1	1	0
		0	0	1	0

Bitwise

AND

$P_1 \rightarrow 1001$

$P_2 \rightarrow 0001$

0001

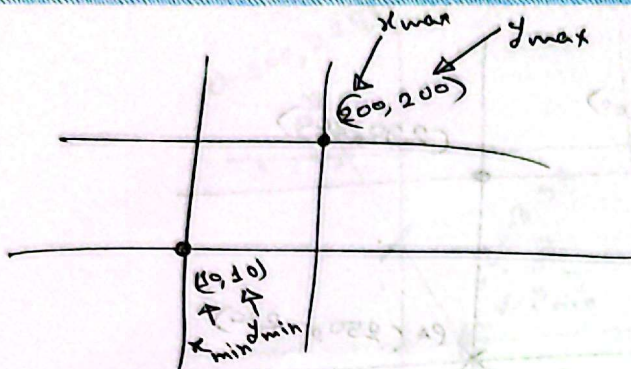
→ Left (L)

completely outside

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /



$$(-10, -10) \quad \text{TBRL} \quad \begin{cases} x_1 < x_{\min} \\ y_1 < y_{\min} \end{cases}$$

$$P_1 \approx 0001 = 0101$$

$$(100, 100) \quad \text{TBRL}$$

$$P_2 \approx 0002 = 1000$$

AND

$$P_1 \rightarrow 0101$$

$$P_2 \rightarrow 1000$$

0000 ← Partially Inside

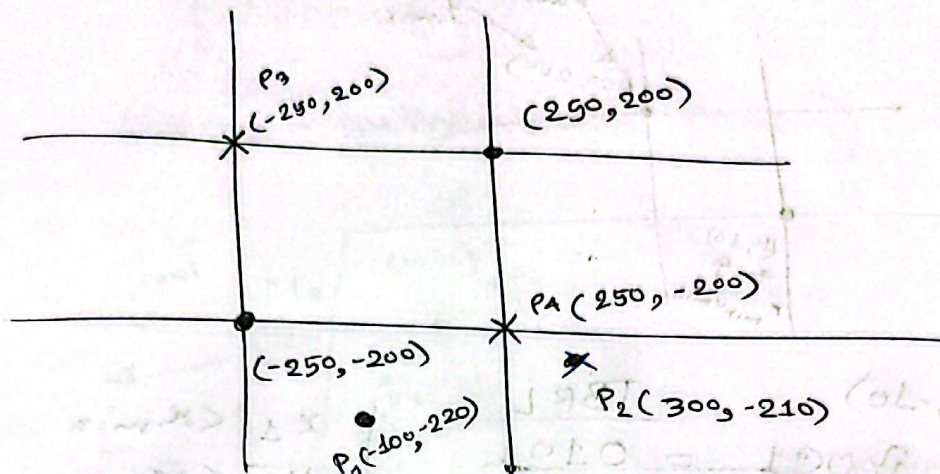
Two points must be 0000 to be Fully Inside.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /



0100

0110

TBRL

$$P_1 \approx 0C1 = 0100$$

$$P_2 \approx 0C2 = 0110$$

$$\text{AND} = 0100$$

object is so, completely outside.

NO clipping

$$P_3 \approx 0C3 = 0000$$

$$P_4 \approx 0C4 = 0000$$

$$\text{AND} = 0000$$

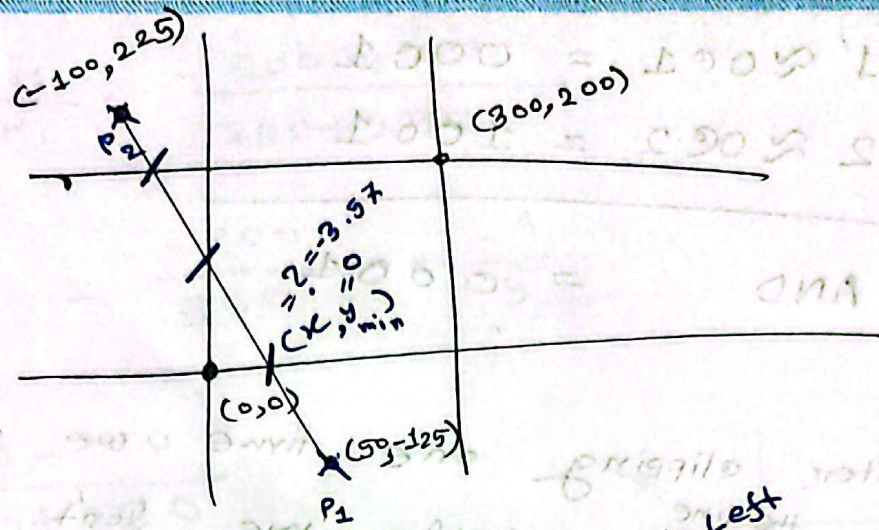
so, completely Inside

No clipping.

TOPIC NAME: _____

DAY: _____

TIME: _____ DATE: / /



$$P_2 \text{ ZOC} = \begin{array}{c} \text{TBRL} \\ \hline 1 \ 0 \ 0 \ 1 \end{array}$$

$$P_1 \text{ ZOC} = \begin{array}{c} 0 \ 1 \ 0 \ 0 \end{array}$$

AND

So, partially Inside
but, we need to clip

$$m = \frac{y_{\min} - y_1}{x - x_1}$$

$$\Rightarrow -\frac{7}{3} = \frac{0 - (-125)}{x - 50}$$

$$\Rightarrow x = -3.57$$

$$(-3.57, 0) \rightarrow \begin{array}{c} \text{TBRL} \\ \hline 0 \ 0 \ 0 \ 1 \end{array}$$

$$\text{slope, } m = \frac{225 + 125}{-100 - 50}$$

$$= \frac{350}{-150}$$

$$= -\frac{7}{3}$$

TOPIC NAME : _____

DAY : _____

TIME : _____

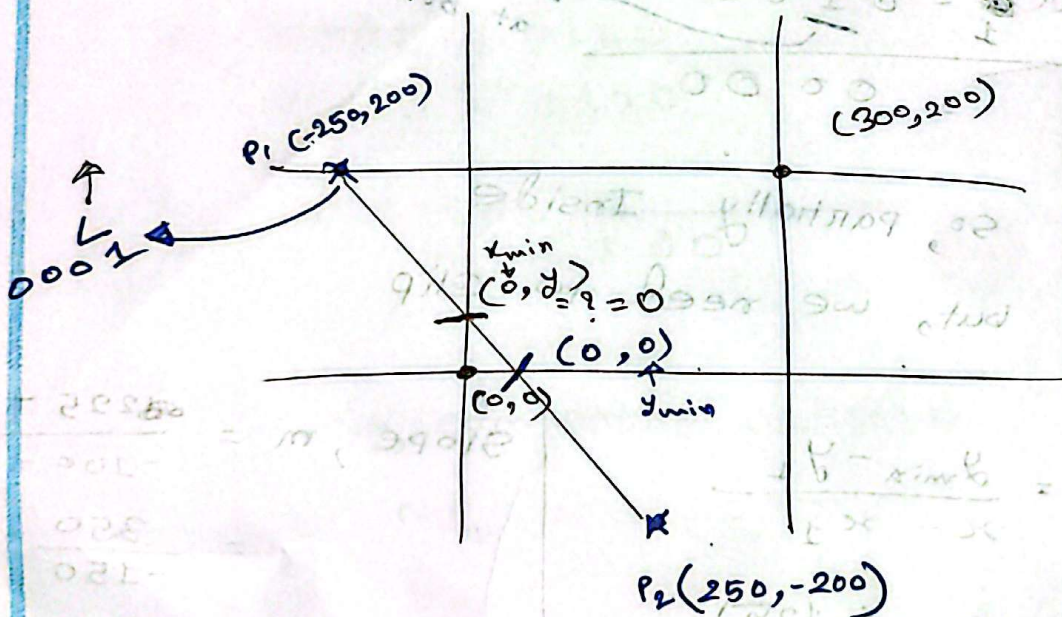
DATE : / /

$$P1' \approx 0001 = 0001$$

$$P2 \approx 0002 = 1001$$

$$\text{AND} = 0001$$

After clipping one time we can see that ~~the~~ line is outside. we don't need to do clip again.



$$P1 \approx 0001 = \begin{matrix} T & B & R & L \\ 0 & 0 & 0 & 1 \end{matrix}$$

$$P2 \approx 0002 = \begin{matrix} 0 & 1 & 0 & 0 \end{matrix}$$

$$\text{AND} = 0000$$

we clip left first

Both are not zero but bitwise

AND is 0000, so partially Inside.

Then Bottom

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

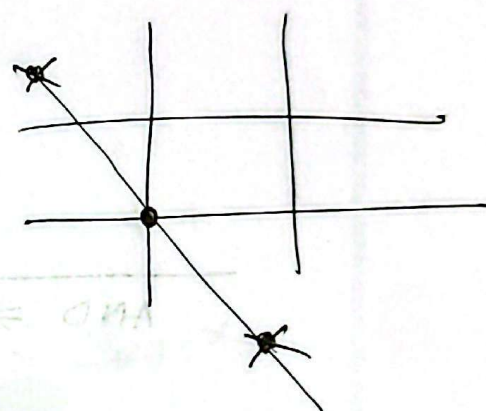
L intersection

$$m = \frac{y - y_1}{x - x_1} = \frac{-200 - 200}{250 + 250}$$

$$= \frac{-400}{500} = -\frac{4}{5}$$

$$-\frac{4}{5} = \frac{-200}{0 + 250}$$

$$\Rightarrow y = 0$$



$$P_1 \approx 0001 = 0000$$

$$P_2 \approx 0002 = 0100$$

AND

$$= 0000$$

} still partially Inside

B intersection

$$\Rightarrow m = \frac{y_{min} - y_2}{x - x_2}$$

$$-\frac{4}{5} = \frac{0 + 200}{x - 250}$$

$$\Rightarrow x = 0$$

$$P_1' \approx 0001 = 0000$$

$$P_2' \approx 0002 = 0000$$

$$AND = 0000$$

} completely Inside

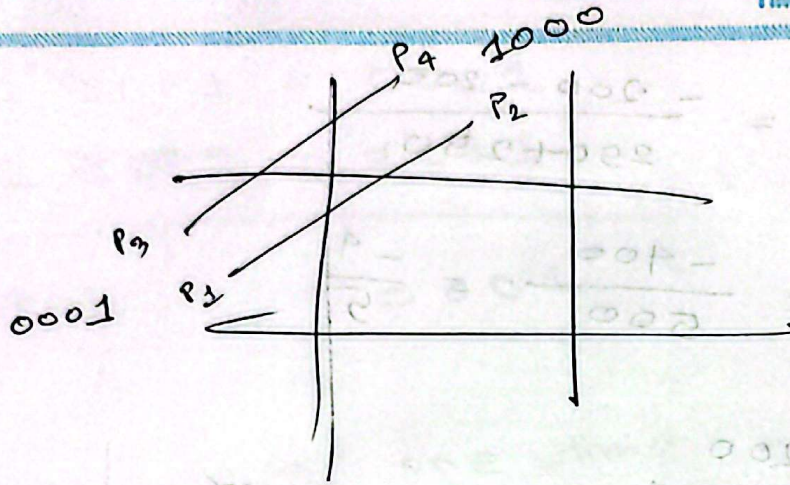
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TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /



0001
1000

AND $\Rightarrow 0000$

✓ we need to clip $P_1 = 1000$

✓ we can't decide which one is

out of range

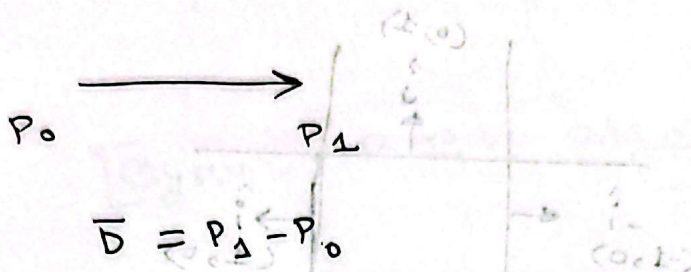
TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : 10 / 07 / 2025

[Parametric equation of Line]

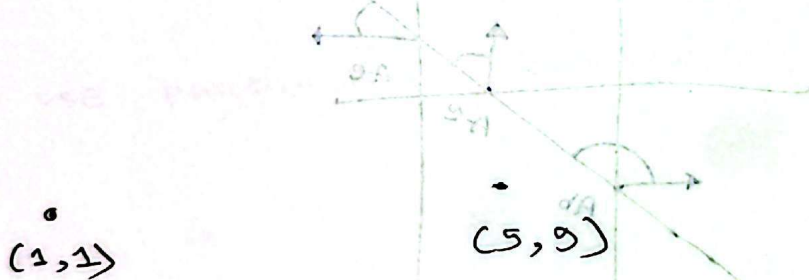


$$\vec{D} = P_1 - P_0$$

$$\Rightarrow P_1 = P_0 + \vec{D}$$

$$\Rightarrow P_1(t) = P_0 + (t) \vec{D}$$

$$P(t) = P_0 + t \cdot (P_1 - P_0)$$



$$P(t) = P_0 + t(P_1 - P_0)$$

$$= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0)$$

$$= (1, 1) + t(5 - 1, 9 - 1)$$

$$= (1, 1) + t(4, 8)$$

$$= (1 + 4t, 1 + 8t)$$

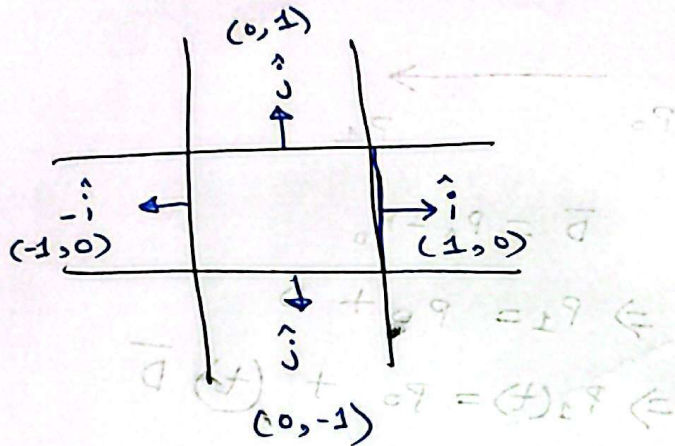
TOPIC NAME : _____

DAY : _____

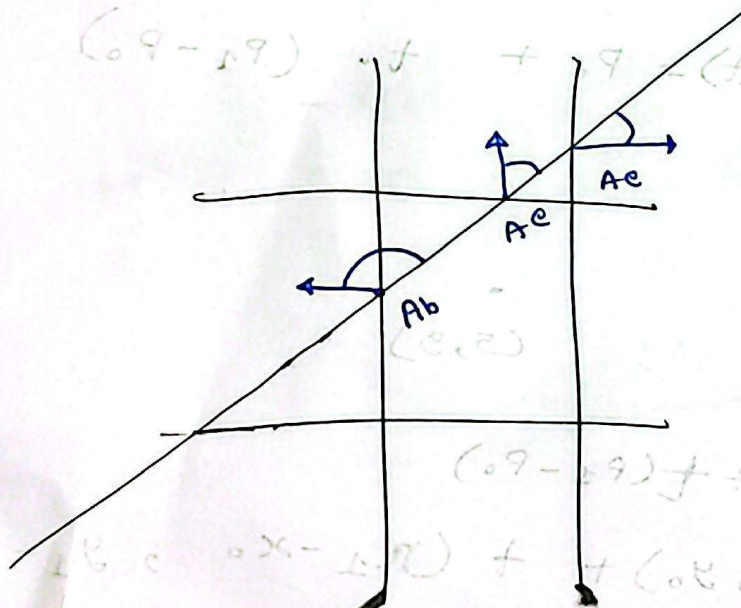
TIME : _____

DATE : / /

normal vectors to boundary



Acute
 $0^\circ < \theta < 90^\circ$
 Obtuse
 $90^\circ < \theta < 180^\circ$



$$N_i \cdot D < 0 \Rightarrow PE$$

angle $> 90^\circ$

Potentially
 Entering

$$N_i \cdot D > 0 \Rightarrow PL$$

angle $< 90^\circ$

Potential
 Leaving



TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$t_E > t_L$; outside from the window

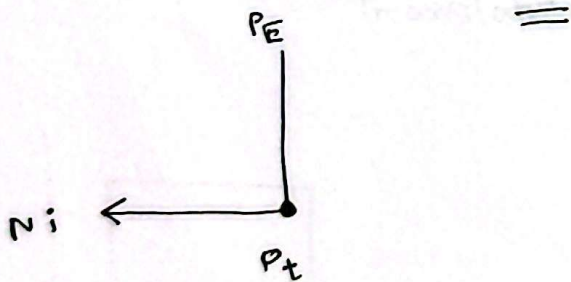
[Cyrus - Beek algo]

✓ calculate t values

✓ PE / PL

✓ choose PE with the highest t
PL " " lowest t

✓ use parametric line eqn.



$$N_i \cdot P(t) - P_{E_i} = 0$$

$$\Rightarrow N_i [P_0 + (P_1 - P_0)t - P_{E_i}] = 0$$

$$\Rightarrow t = \frac{N_i \cdot [P_0 - P_{E_i}]}{-N_i (P_1 - P_0)}$$

TOPIC NAME : _____

DAY : _____

TIME : _____

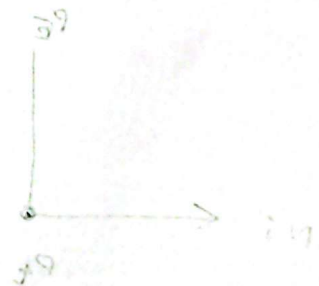
DATE : / /

$$t_{left} = \frac{-(x_0 - x_{min})}{(x_1 - x_0)}$$

$$t_{right} = \frac{-(x_0 - x_{max})}{(x_1 - x_0)}$$

$$t_{top} = \frac{-(y_0 - y_{max})}{(y_1 - y_0)}$$

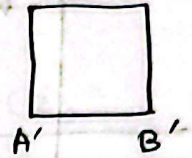
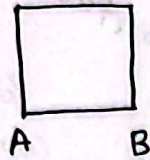
$$t_{bottom} = \frac{-(y_0 - y_{min})}{(y_1 - y_0)}$$



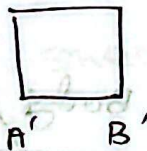
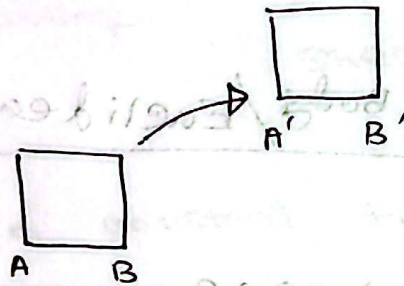
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DAY : _____

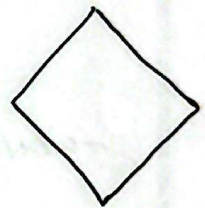
TIME : _____ DATE : / /



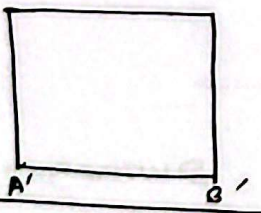
Identity



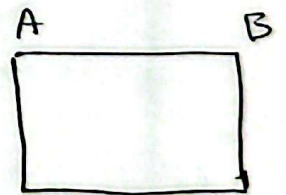
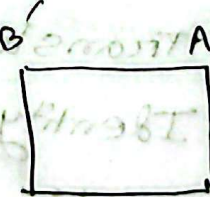
Translation



Rotation



Scaling



Reflection

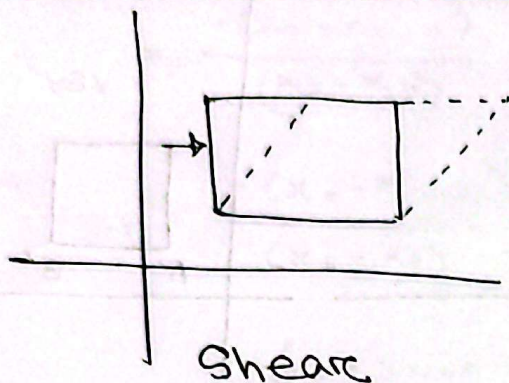
Uniform / Isotropic

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /



distorted

Rigid Body / Euclidean Transform

1) preserve distances

2) " angles

Translation
Identity
Rotation

Similitudes / Similarity Transforms

1) preserve angles

Isotropic
Scaling

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Linear Transformations

$$f(x) = 2x$$

$$\begin{aligned} \textcircled{5} \quad f(a+b) &= f(a) + f(b) \\ \Rightarrow f(2+3) &= f(2) + f(3) \\ \Rightarrow 10 &= 10 \end{aligned}$$

$$f(cx) = cf(x)$$

$$\Rightarrow f(2 \times 2) = 2f(2)$$

$$\Rightarrow 4 = 2 \times 2 = 4$$

Identity
Relation

Isotropic
Scaling

Scaling
Reflection
Shear

If I add a constant term, Translation
Don't maintain linear transformation.

Affine Transformations

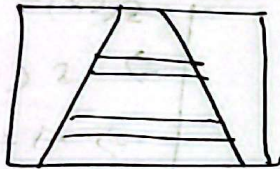
1) preserve parallel lines.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Projective Transformations



P_1

$$\begin{aligned} P_2 &= S_2(P_1) \\ P_3 &= S_2(P_2) \\ P' &= S_3(P_3) \end{aligned}$$

↓
matrix

$\Delta = P'$

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

Function takes more time
so it's better to use matrix.

$$x' = ax + by + c$$

$$y' = dx + ey + f$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} c \\ f \end{bmatrix}$$

Cartesian Formula

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Formulation

$A'(x + \Delta x, y + \Delta y)$

$A(x, y)$

$$x' = x + \Delta x$$

$$y' = y + \Delta y$$

$$\begin{bmatrix} 1 & 0 & \Delta x \\ 0 & 1 & \Delta y \\ 0 & 0 & 1 \end{bmatrix}$$

TOPIC NAME: _____

DAY: _____

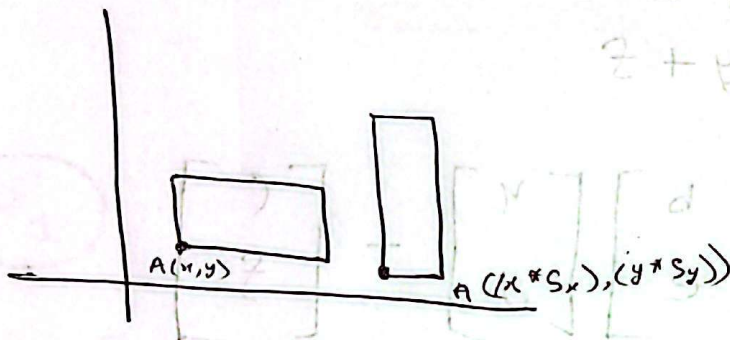
TIME: _____

DATE: / /

2D $\rightarrow 3 \times 3$

3D $\rightarrow 4 \times 4$

[Scaling]



~~Scaling factor~~

Scaling factor $\rightarrow S_x, S_y, S_z$

$S_x = S_y \rightarrow$ Differential

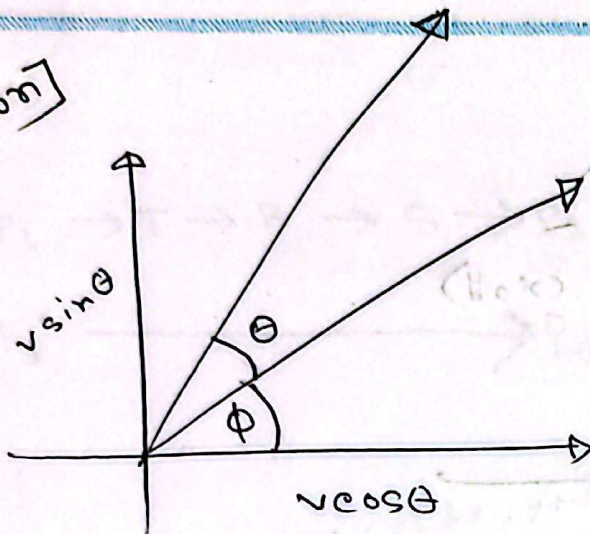
$S_x = S_y \rightarrow$ uniform

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

[Rotation]

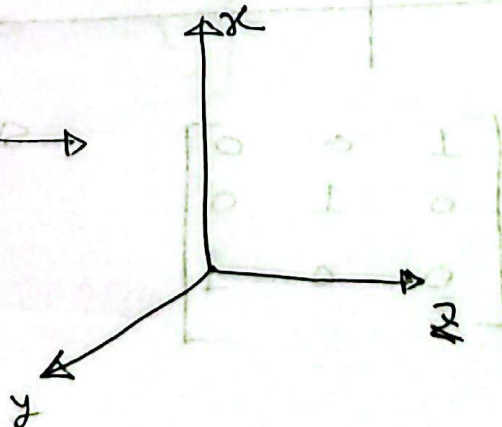
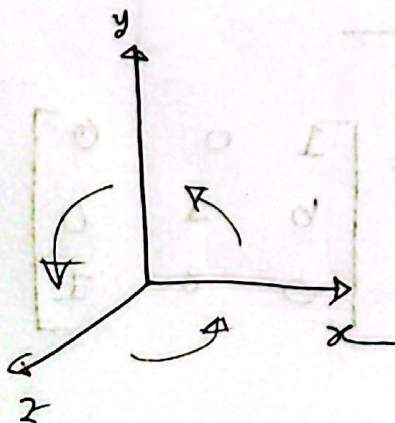


$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$x' = x \cos\theta - y \sin\theta$$

$$y' = x \sin\theta + y \cos\theta$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



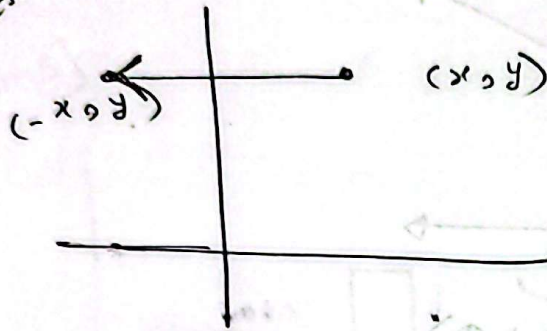
TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

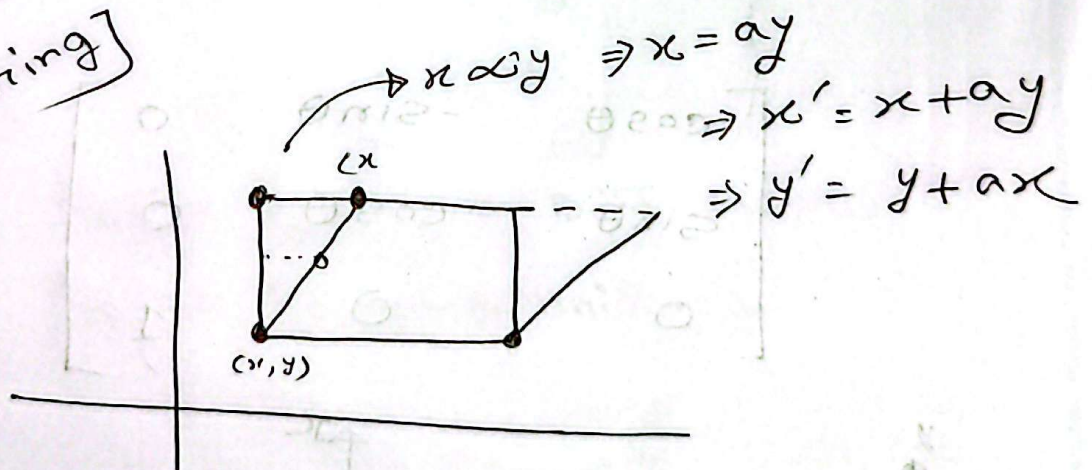
Mirror Reflection



3d

(x, y, z) plane

Shearing



$$Sh_x = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Sh_y = \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

TOPIC NAME : _____

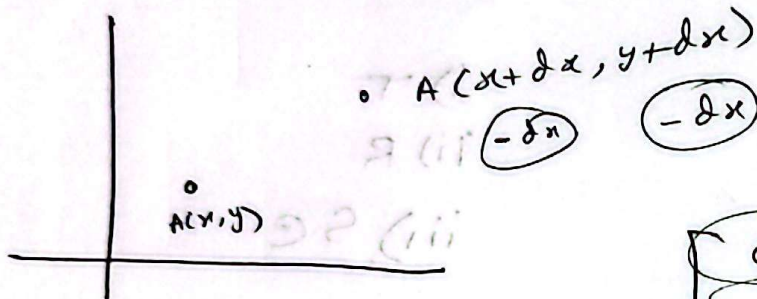
DAY : _____

TIME : _____ DATE : / /

[Inverse]

$$P_1 \rightarrow T \rightarrow R \rightarrow S \rightarrow P_1'$$

$$P_1' \rightarrow P_1$$



$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(-\theta) & -\sin(-\theta) & 0 \\ \sin(-\theta) & \cos(-\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

TOPIC NAME: _____

DAY: _____

TIME: _____

DATE: / /

[Composing Transformations]

$$A \times B \neq B \times A \longrightarrow \text{Order matters}$$

$$A(BC) = AB \circ$$

$$P_1 =$$

$$P_1' = \begin{bmatrix} M_{sc} & M_R & M_T & P_1 \end{bmatrix}$$

i) T

ii) R

iii) S

$$P_1'' = M_T M_R M_{sc} P_1$$